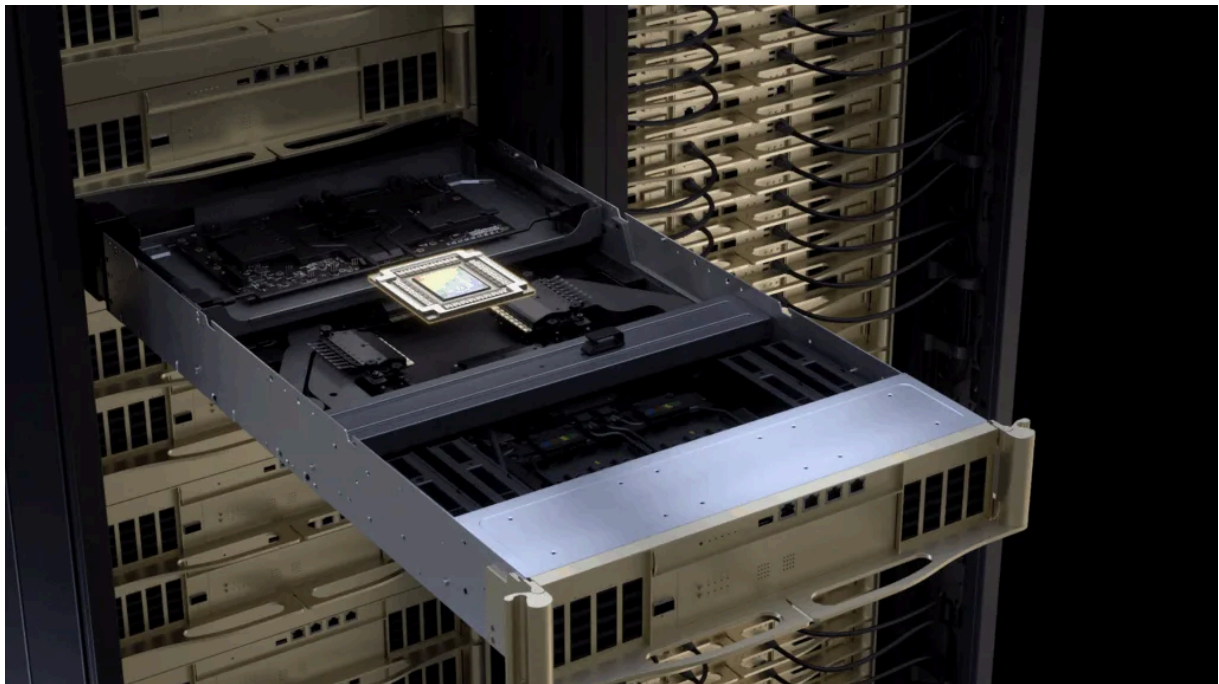




Scaling Power-Efficient AI Factories with NVIDIA Spectrum-X Ethernet Photonics



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By [Ashkan Seyed](#)

AI-Generated Summary



NVIDIA is bringing the world's first optimized Ethernet networking with co-packaged optics to AI factories, enabling scale-out and scale-across on the NVIDIA Rubin platform with NVIDIA Spectrum-X Ethernet Photonics, the flagship switch for multi-trillion-parameter AI infrastructure.

This blog post explores key optimizations and innovations in the protocol and hardware of Spectrum-X Ethernet Photonics that enable power-efficient, reliable,

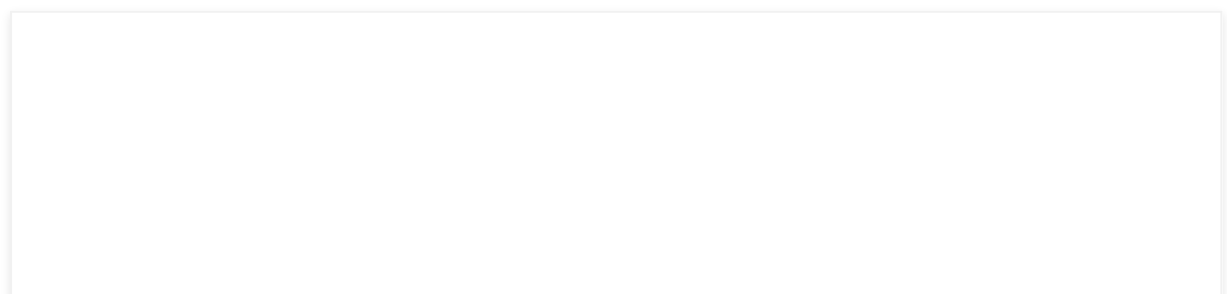
and resilient co-packaged optical networks for giga-scale AI factories.

How Ethernet for AI enables scalable training and inference on the NVIDIA Rubin Platform

Ultra-low-jitter Ethernet networking plays a vital role in scaling AI factories, as it ensures consistent and reliable data transmission across the entire infrastructure. By minimizing jitter, AI systems can achieve efficient token throughput regardless of batch size, which is crucial for handling diverse and demanding workloads. This ability supports seamless multi-tenancy within a single AI factory, for multiple users and applications to operate concurrently without performance degradation.

It also improves the dispatch efficiency of models based on the Mixture of Experts (MoE) architecture, enabling faster expert selection and improved overall model performance, as shown in Figure 1. As a result, AI factories can operate at greater speed, reliability, and scalability.

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compared to off-the-shelf Ethernet solutions, ensuring AI workloads run

How to Connect Distributed Data Centers Into Large AI Factories with Scale-Across Networking

Scaling AI Factories with Co-Packaged Optics for Better Power Efficiency

A New Era in Data Center Networking with NVIDIA Silicon Photonics-based Network Switching

These combined innovations equip AI factories with robust, scalable network infrastructure capable of supporting next-generation artificial intelligence applications.

Powering Next-Generation AI Networking with NVIDIA SuperNICs

Ethernet Photonics switches deliver ultra-low jitter networking for AI factories to grow in speed, reliability, and scalability, and establish robust infrastructure for next-generation applications. For more information, see the [NVIDIA Silicon Photonics](#) page.

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About Ashkan Seyedi

Ashkan Seyedi is a director of product marketing who works on developing high-bandwidth, efficient optical interconnects for AI and high-performance computing. He previously worked at Intel and Hewlett Packard Enterprise. Ashkan received a dual bachelor's in electrical and computer engineering from the University of Missouri in Columbia and a Ph.D. from the University of Southern California, where he worked on photonic crystal devices, high-speed nanowire photodetectors, efficient white LEDs, and solar cells.

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